

Chenmu Zhang

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EDUCATION

- **University of Texas at Austin** 2018 – 2023
Ph.D. in Materials Engineering
Advisor: Prof. Yuanyue Liu
Thesis: “Phonon-limited Carrier Transport in 2D Materials from First Principles”
- **University of Science and Technology of China (USTC)** 2014 – 2018
B.S. in Physics
Advisor: Prof. Wenguang Zhu
Thesis: “First Principles Calculations on 2D Ferroelectric Materials under Uniaxial Strain”

WORK EXPERIENCES

- **Research Scientist** 2026 – Present
Rice University, Houston, TX, United States
Developed autonomous code-first LLM agents that orchestrate multi-code simulation workflows on HPC clusters for end-to-end materials exploration
- **Postdoctoral Research Associate** 2023 – 2026
Rice University, Houston, TX, United States
Developed and utilized first-principles calculations to study the dynamic process of defect in transition metal dichalcogenide monolayers
- **Graduate Research Assistant** 2018 – 2023
University of Texas at Austin, Austin, TX, United States
Improved and developed first-principles algorithms for phonon-limited mobility of 2D semiconductors
Developed Monte Carlo code for high-field electron transport in 2D semiconductors
Studied on other electron-phonon scattering factors in 2D materials including remote phonon and substrate effect

PUBLICATIONS

1. **Zhang, C.**; Yakobson, B. I. MatClaw: An Autonomous Code-First LLM Agent for End-to-End Materials Exploration. arXiv preprint arXiv:2604.02688 **2026**.
2. **Zhang, C.**; Liu, Y. Anomalous enhancement of carrier mobility by remote phonons. arXiv preprint arXiv:2404.08114 **2025**.
3. **Zhang, C.**; Liu, Y. Two-dimensional Semiconductor Computational Carrier Mobility Genome. arXiv preprint arXiv:2404.08117 **2024**.
4. **Zhang, C.**; Liu, Y. Electron-Surface Scattering from First-Principles. *ACS Nano* **2024**, 18(40), 27433–27439.
5. **Zhang, C.**; Xiao, Z.; Paddock, R.; Cullinan, M.; Tehrani, M.; Liu, Y. Effects of Graphene Doping on the Electrical Conductivity of Copper. *Advanced Functional Materials* **2024**, 34(45), 2407569.
6. **Zhang, C.**; Wang, R.; Mishra, H.; Liu, Y. Two-Dimensional Semiconductors with High Intrinsic Carrier Mobility at Room Temperature. *Physical Review Letters* **2023**, 130(8), 087001.
7. **Zhang, C.**; Liu, Y. Phonon-limited transport of two-dimensional semiconductors: Quadrupole scattering and free carrier screening. *Physical Review B* **2022**, 106(11), 115423.

8. **Zhang, C.**; Cheng, L.; Liu, Y. Understanding high-field electron transport properties and strain effects of monolayer transition metal dichalcogenides. *Physical Review B* **2020**, 102(11), 115405.
9. **Zhang, C.**; Cheng, L.; Liu, Y. Role of flexural phonons in carrier mobility of two-dimensional semiconductors: free standing vs on substrate. *Journal of Physics: Condensed Matter* **2021**, 33(23), 234003.
10. Xiao, Z.; Guo, R.; **Zhang, C.**; Liu, Y. Point Defect Limited Carrier Mobility in 2D Transition Metal Dichalcogenides. *ACS Nano* **2024**, 18(11), 8511–8516.
11. Cheng, L.; **Zhang, C.**; Liu, Y. Intrinsic charge carrier mobility of 2D semiconductors. *Computational Materials Science* **2021**, 194, 110468.
12. Cheng, L.; **Zhang, C.**; Liu, Y. Why Two-Dimensional Semiconductors Generally Have Low Electron Mobility. *Physical Review Letters* **2020**, 125(17), 177701.
13. Cheng, L.; **Zhang, C.**; Liu, Y. The Optimal Electronic Structure for High-Mobility 2D Semiconductors: Exceptionally High Hole Mobility in 2D Antimony. *Journal of the American Chemical Society* **2019**, 141(41), 16296–16302.
14. Cheng, L.; **Zhang, C.**; Liu, Y. How to resolve a phonon-associated property into contributions of basic phonon modes. *Journal of Physics: Materials* **2019**, 2(4), 045005.

TEACHING

- Teaching assistant in Structure of Materials, fall, 2022
- Teaching assistant in Structure of Materials, fall, 2021